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[162060] LOW-RESOURCE MACHINE TRANSLATION  
SoSe 2026

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| <b>LANGUAGE:</b> | English              | <b>TEACHER:</b>  | Christian Schuler           |
| <b>PLACE:</b>    | C7.2 - 1.05          | <b>OFFICE:</b>   | D3 1 (Altbau) +1, Room 1.09 |
| <b>TIME:</b>     | 16:15-17:45          | <b>E-MAIL:</b>   | christian.schuler@dfki.de   |
| <b>START:</b>    | 14.04.2026           | <b>OBERBABO:</b> | Simon Ostermann             |
| <b>CREDITS:</b>  | 4 CP (R), 7 CP (R+H) |                  |                             |
| <b>WORKLOAD:</b> | 2 SWS                |                  |                             |

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### COURSE DESCRIPTION

This seminar explores the challenges and possibilities of machine translation in contexts where data is scarce, incomplete, or entirely absent. Rather than focusing on standard, high-resource scenarios, the course examines what happens when core assumptions of modern NLP break down: when languages lack large corpora, established tools, or even a written tradition. Students will engage with both the technical and conceptual foundations of low-resource machine translation. Topics include data acquisition and corpus creation, evaluation under uncertainty, multilingual and transfer learning approaches, and strategies such as data augmentation, zero- and few-shot learning, and active learning. Alongside these methods, the course highlights the role of linguistic knowledge as a structured alternative to purely data-driven approaches. A central thread throughout the seminar is the critical examination of the broader context in which these systems are developed and deployed. Questions of ethics, bias, and responsibility are treated as integral to the field: whose languages are supported, how linguistic communities are represented, and what risks arise when systems are applied in sensitive or high-stakes domains. The course is intended for students with an interest in natural language processing, machine learning, and linguistic diversity, who want to understand both the limitations of current approaches and the strategies being developed to overcome them. It is particularly suited for those interested in research questions at the intersection of technology, language, and society.

### SEMINAR, SEMINAR- WHAT EVEN IS A SEMINAR?

“A seminar is a form of (...) recurring meetings, focusing each time on some particular subject, in which everyone present is requested to participate. This is often accomplished through an ongoing Socratic dialogue (...) It is essentially a place where assigned readings are discussed, questions can be raised and debates can be conducted.” \*

### SEMINAR SCHEDULE

Each session has a pool of papers, and we plan to have two presentations per session. Students present a paper of their choice (under the constraint that we need to fill each session) from the session's paper pool and lead the class discussion.

Some words on grading: This seminar is meant to be as interactive as possible. Final grades will be based on students' presentations, term papers (optional), but also on participation and discussion in class.

The participants are expected to prepare for classes accordingly, by reading the relevant papers and also doing background reading, if necessary. Based on this preparation, the participants should be able to discuss the presented papers in depth and to understand relevant context during the discussion.

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\*<https://en.wikipedia.org/wiki/Seminar>

## INTRODUCTION, COURSE OVERVIEW, AND ADMINISTRATIVE MATTERS

*“What are we going to do here?”*

| Session                                 | Date       | Content                                    | Paper                   | Student |
|-----------------------------------------|------------|--------------------------------------------|-------------------------|---------|
| Phase 1 - Warm-Up                       |            |                                            |                         |         |
| 1                                       | 14.04.2026 | Administrative Session & Paper Pool        |                         |         |
| 2                                       | 21.04.2026 | Introductory Lecture                       |                         |         |
| 3                                       | 28.04.2026 | Lingering Questions (optional)             |                         |         |
| 4                                       | 05.05.2026 | Lingering Questions (optional)             |                         |         |
| 5                                       | 12.05.2026 | LREC (no meeting)                          |                         |         |
| Phase 2 - Student's Paper Presentations |            |                                            |                         |         |
| 6                                       | 19.05.2026 | Problem Space & Linguistic Foundations     | (Kornai, 2013)          | Q.M.    |
| 7                                       | 26.05.2026 | Problem Space & Linguistic Foundations     | (Shompa, 2025)          | P.H.    |
| 8                                       | 02.06.2026 | Data Engineering & Augmentation            | (Riabi et al., 2023)    | A.K.    |
| 9                                       | 09.06.2026 | Data Engineering & Augmentation            | (Riabi et al., 2025)    | J.W.    |
| 10                                      | 16.06.2026 | Ethics & Policy                            |                         | A?      |
| 11                                      | 23.06.2026 | Evaluation & Quality Estimation            | (Papineni et al., 2002) | N.G.    |
| 12                                      | 30.06.2026 | Sub-word Modeling & Representation         |                         |         |
| 13                                      | 07.06.2026 | Learning Paradigms & Architectural Methods | (Kholodna et al., 2024) | F.W.    |
| Phase 3 - Wrap-Up                       |            |                                            |                         |         |
| 14                                      | 14.07.2026 | Class Review & Synthesis                   |                         |         |

## PROBLEM SPACE & LINGUISTIC FOUNDATIONS

*“What defines low-resource languages and why do they pose unique challenges for MT?”*

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Digital Language Death

Kornai (2013) | [URL](#)

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Survey of Low-Resource Machine Translation

Haddow et al. (2022) | [URL](#)

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Speech Translation and the End-to-End Promise: Taking Stock of Where We Are

Sperber and Paulik (2020) | [URL](#)

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The State and Fate of Linguistic Diversity and Inclusion in the NLP World

Joshi et al. (2020) | [URL](#)

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ELLORA: Enabling Low Resource Languages with Technology

Bali et al. (2019) | [URL](#)

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Cyber-geography of languages: Part 1: method, results and focus on English

Pimienta and de Oliveira (2022) | [URL](#)

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### Additional Reading

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Language Death

Crystal (2000) | [URL](#)

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Atlas of the world's languages in danger

Moseley and Nicolas (2010) | [URL](#)

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## ETHICS & POLICY

*“Who benefits, who is harmed, and what governance is required?”*

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On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?

Bender et al. (2021) | [URL](#)

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Challenges of Language Technologies for the Indigenous Languages of the Americas

Mager et al. (2018) | [URL](#)

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Local Languages, Third Spaces, and other High-Resource Scenarios

Bird (2022) | [URL](#)

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The State and Fate of Linguistic Diversity and Inclusion in the NLP World

Joshi et al. (2020) | [URL](#)

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Decolonising Speech and Language Technology

Bird (2020) | [URL](#)

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Not always about you: Prioritizing community needs when developing endangered language technology

Liu et al. (2022) | [URL](#)

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### Additional Reading

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## DATA ENGINEERING & AUGMENTATION

*“How to acquire, curate, or synthesize data under extreme scarcity?”*

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Europarl: A Parallel Corpus for Statistical Machine Translation

Koehn (2005) | [URL](#)

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CCMatrix: Mining Billions of High-Quality Parallel Sentences on the Web

Schwenk et al. (2021) | [URL](#)

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Findings of the WMT 2018 Shared Task on Parallel Corpus Filtering

Koehn et al. (2018) | [URL](#)

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Seven Dimensions of Portability for Language Documentation and Description

Bird and Simons (2002) | [URL](#)

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Massively Multilingual Sentence Embeddings for Zero-Shot Cross-Lingual Transfer and Beyond

Artetxe and Schwenk (2019) | [URL](#)

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Improving Parallel Sentence Mining for Low-Resource and Endangered Languages

Okabe et al. (2025) | [URL](#)

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Parallel Data, Tools and Interfaces in OPUS

Tiedemann (2012) | [URL](#)

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The OPUS-MT Dashboard: A Toolkit for a Systematic Evaluation of Open Machine Translation Models

Tiedemann and de Gibert (2023) | [URL](#)

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Improving Neural Machine Translation Models with Monolingual Data

Sennrich et al. (2016a) | [URL](#)

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Understanding Back-Translation at Scale

Edunov et al. (2018) | [URL](#)

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Exploring Diversity in Back Translation for Low-Resource NMT

Burchell et al. (2022) | [URL](#)

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Beyond Dataset Creation: Critical View of Annotation Variation and Bias Probing of a Dataset for Online Radical Content Detection

Riabi et al. (2025) | [URL](#)

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Treebanking user-generated content: a UD based overview of guidelines, corpora and unified recommendations

Sanguinetti et al. (2023) | [URL](#)

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Enriching the NArabizi Treebank: A Multifaceted Approach to Supporting an Under-Resourced Language

Riabi et al. (2023) | [URL](#)

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## Additional Reading

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## SUB-WORD MODELING & REPRESENTATION

*“How to represent language when standard tokenization fails?”*

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SentencePiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing  
Kudo and Richardson (2018) | [URL](#)

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Unsupervised Multilingual Learning for POS Tagging  
Snyder et al. (2008) | [URL](#)

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Unsupervised Learning of the Morphology of a Natural Language  
Goldsmith (2001) | [URL](#)

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Neural Machine Translation of Rare Words with Subword Units  
Sennrich et al. (2016b) | [URL](#)

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Modeling Language Variation and Universals: A Survey on Typological Linguistics for Natural Language Processing  
Ponti et al. (2019) | [URL](#)

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Saliency-driven Word Alignment Interpretation for Neural Machine Translation  
Ding et al. (2019) | [URL](#)

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Selective Sharing for Multilingual Dependency Parsing  
Naseem et al. (2012) | [URL](#)

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Neural Machine Translation of Rare Words with Subword Units  
Sennrich et al. (2016b) | [URL](#)

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LINSPECTOR: Multilingual Probing Tasks for Word Representations  
Şahin et al. (2020) | [URL](#)

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Zero-Shot Neural Machine Translation with Self-Learning Cycle  
Lakew et al. (2021) | [URL](#)

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A Rule-Based Kurdish Text Transliteration System  
Ahmadi (2019) | [URL](#)

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Measuring Similarity by Linguistic Features rather than Frequency  
Delmonte and Busetto (2022) | [URL](#)

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### **Additional Reading**

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## **EVALUATION & QUALITY ESTIMATION**

*“How to measure MT quality reliably when references and benchmarks are scarce?”*

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BLEU: A Method for Automatic Evaluation of Machine Translation  
Papineni et al. (2002) | [URL](#)

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A Structured Review of the Validity of BLEU  
Reiter (2018) | [URL](#)

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BLEURT: Learning Robust Metrics for Text Generation  
Sellam et al. (2020) | [URL](#)

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COMET: A Neural Framework for MT Evaluation  
Rei et al. (2020) | [URL](#)

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### **Additional Reading**

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## DOMAIN ADAPTATION & MOADLITIES

*“What happens to machine translation when the language was never primarily a text?”*

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Analyzing phonetic variation in the traditional English dialects: Simultaneously clustering dialects and phonetic features

Wieling et al. (2013) | [URL](#)

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### Additional Reading

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## LEARNING PARADIGMS & ARCHITECTURAL METHODS

*“How to train or adapt models effectively with minimal supervision?”*

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Google’s Multilingual Neural Machine Translation System: Enabling Zero-Shot Translation

Johnson et al. (2017) | [URL](#)

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Massively Multilingual Neural Machine Translation

Aharoni et al. (2019) | [URL](#)

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Unsupervised Cross-lingual Representation Learning at Scale

Conneau et al. (2020) | [URL](#)

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Transfer Learning across Low-Resource, Related Languages for Neural Machine Translation

Nguyen and Chiang (2017) | [URL](#)

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Polylingual Topic Models

Mimno et al. (2009) | [URL](#)

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Ensemble Learning for Multi-source Neural Machine Translation

Garmash and Monz (2016) | [URL](#)

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Attention is All You Need

Vaswani et al. (2017) | [URL](#)

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Neural Machine Translation

Koehn (2020) | [URL](#)

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Unsupervised Neural Machine Translation

Artetxe et al. (2018) | [URL](#)

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Neural Machine Translation through Active Learning on Low-Resource Languages: The Case of Spanish to Mapudungun

Pendas et al. (2023) | [URL](#)

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LLMs in the Loop: Leveraging Large Language Model Annotations for Active Learning in Low-Resource Languages

Kholodna et al. (2024) | [URL](#)

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Zero-Shot Neural Machine Translation with Self-Learning Cycle

Lakew et al. (2021) | [URL](#)

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Ensemble Learning for Multi-source Neural Machine Translation  
Garmash and Monz (2016) | [URL](#)

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A Survey of Domain Adaptation for Neural Machine Translation  
Chu and Wang (2018) | [URL](#)

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Finding the Right Recipe for Low Resource Domain Adaptation in Neural Machine Translation  
Adams et al. (2022) | [URL](#)

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On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?  
Bender et al. (2021) | [URL](#)

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### Additional Reading

No Language Left Behind: Scaling Human-Centered Machine Translation  
Team et al. (2022) | [URL](#)

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## EMERGING DIRECTIONS & OPEN PROBLEMS

*“What new paradigms and unresolved issues will shape the near future?”*

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Should All Cross-Lingual Embeddings Speak English?  
Anastasopoulos and Neubig (2020) | [URL](#)

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Natural Language Processing for Dialects of a Language: A Survey  
Joshi et al. (2025) | [URL](#)

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Large Multimodal Models for Low-Resource Languages: A Survey  
Lupascu et al. (2025) | [URL](#)

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Neural Sign Language Translation  
Camgoz et al. (2018) | [URL](#)

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The Tatoeba Translation Challenge - Realistic Data Sets for Low Resource and Multilingual MT  
Tiedemann (2020) | [URL](#)

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Low-Resource Speech Translation  
Bansal (2019) | [URL](#)

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Sign Language Translation: A Survey of Approaches and Techniques  
Liang et al. (2023) | [URL](#)

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Speech Translation and the End-to-End Promise: Taking Stock of Where We Are  
Sperber and Paulik (2020) | [URL](#)

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A Practical Tool to Help Automate Interlinear Glossing: a Study on Mukr Kurdish  
Asadpour et al. (2025) | [URL](#)

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Learning to Recognize Dialect Features  
Demszky et al. (2021) | [URL](#)

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Multi-VALUE: A Framework for Cross-Dialectal English NLP  
Ziems et al. (2023) | [URL](#)

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Translation Asymmetry in LLMs as a Data Augmentation Factor: A Case Study for 6 Romansh Language Varieties  
Vamvas et al. (2026) | [URL](#)

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*Versteasch du mi?* Computational and Socio-Linguistic Perspectives on GenAI, LLMs, and Non-Standard Language  
Platzgummer et al. (2026) | [URL](#)

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Can Character-based Language Models Improve Downstream Task Performances In Low-Resource And Noisy Language Scenarios?  
Riabi et al. (2021) | [URL](#)

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Can Multilingual Language Models Transfer to an Unseen Dialect? A Case Study on North African Arabizi  
Muller et al. (2020) | [URL](#)

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Large Multimodal Models for Low-Resource Languages: A Survey  
Lupascu et al. (2025) | [URL](#)

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Disentangling meaning from language in LLM-based machine translation  
Lasnier et al. (2026) | [URL](#)

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### **Additional Reading**

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Proceedings of the First Workshop on NLP applications to field linguistics  
Serikov et al. (2022) | [URL](#)

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Proceedings of the Second Workshop on NLP Applications to Field Linguistics  
Serikov et al. (2023) | [URL](#)

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Proceedings of the Third Workshop on NLP Applications to Field Linguistics  
Serikov et al. (2024) | [URL](#)

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Proceedings of the Fourth Workshop on NLP Applications to Field Linguistics  
Le Ferrand et al. (2025) | [URL](#)

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## REFERENCES

- Virginia Adams, Sandeep Subramanian, Mike Chrzanowski, Oleksii Hrinchuk, and Oleksii Kuchaiev. 2022. Finding the Right Recipe for Low Resource Domain Adaptation in Neural Machine Translation. *arXiv:2206.01137* [cs] [doi:10.48550/arXiv.2206.01137](https://doi.org/10.48550/arXiv.2206.01137) 7
- Roei Aharoni, Melvin Johnson, and Orhan Firat. 2019. Massively Multilingual Neural Machine Translation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Jill Burstein, Christy Doran, and Thamar Solorio (Eds.). Association for Computational Linguistics, Minneapolis, Minnesota, 3874–3884. [doi:10.18653/v1/N19-1388](https://doi.org/10.18653/v1/N19-1388) 6
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- Hiwa Asadpour, Shu Okabe, and Alexander Fraser. 2025. A Practical Tool to Help Automate Interlinear Glossing: A Study on Mukri Kurdish. In *Proceedings of the Fourth Workshop on NLP Applications to Field Linguistics*, Éric Le Ferrand, Elena Klyachko, Anna Postnikova, Tatiana Shavrina, Oleg Serikov, Ekaterina Voloshina, and Ekaterina Vylomova (Eds.). Association for Computational Linguistics, Vienna, Austria, 65–75. <https://aclanthology.org/2025.fieldmatters-1.6/> 7
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- Sameer Bansal. 2019. Low-Resource Speech Translation. (Dec. 2019). [doi:10.7488/era/86](https://doi.org/10.7488/era/86) 7
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- Steven Bird. 2022. Local Languages, Third Spaces, and Other High-Resource Scenarios. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 7817–7829. [doi:10.18653/v1/2022.acl-long.539](https://doi.org/10.18653/v1/2022.acl-long.539) 3
- Steven Bird and Gary Simons. 2002. Seven Dimensions of Portability for Language Documentation and Description. *CoRR* cs.CL/0204020 (April 2002). [doi:10.1353/lan.2003.0149](https://doi.org/10.1353/lan.2003.0149) 4

- Laurie Burchell, Alexandra Birch, and Kenneth Heafield. 2022. Exploring Diversity in Back Translation for Low-Resource Machine Translation. In *Proceedings of the Third Workshop on Deep Learning for Low-Resource Natural Language Processing*. 67–79. arXiv:2206.00564 [cs] [doi:10.18653/v1/2022.deeplo-1.8](https://doi.org/10.18653/v1/2022.deeplo-1.8) 4
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- Rodolfo Delmonte and Nicolò Busetto. 2022. Measuring Similarity by Linguistic Features Rather than Frequency. In *Proceedings of the 18th Joint ACL - ISO Workshop on Interoperable Semantic Annotation within LREC2022*, Harry Bunt (Ed.). European Language Resources Association, Marseille, France, 42–52. <https://aclanthology.org/2022.isa-1.6/> 5
- Dorottya Demszky, Devyani Sharma, Jonathan Clark, Vinodkumar Prabhakaran, and Jacob Eisenstein. 2021. Learning to Recognize Dialect Features. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Online, 2315–2338. [doi:10.18653/v1/2021.naacl-main.184](https://doi.org/10.18653/v1/2021.naacl-main.184) 7
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